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A Novel Super-Resolution-Based Multi-Scale Pore Network Modeling Method for Characterizing Complex Rock Inner Structures

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Abstract

Characterizing the complex inner structure of heterogeneous rocks with multi-scale pores presents significant challenges, as conventional imaging techniques cannot simultaneously achieve both high resolution and large field of view (FOV). While high resolution is essential for capturing fine microstructures, a large FOV is required to represent pore heterogeneity, yet these two objectives are fundamentally opposed in imaging systems. To overcome this limitation, we develop an innovative workflow that integrates deep learning-based super-resolution reconstruction with multi-scale pore network modeling (PNM), enabling comprehensive characterization of pore structures across different scales.

Because there is no paired different-resolution images, we use down-sampled images (LR) as training data, and the original high-resolution images (HR) as reference. The workflow begins with high-resolution (2.68 $\mu\text{m}/\text{voxel}$) micro-CT imaging(HR) of a 6 mm core plug, which is downsampled to create low-resolution (10.72 $\mu\text{m}/\text{voxel}$) training images (SR) for a Super-Resolution Generative Adversarial Network (SR-GAN). The trained SR-GAN model then applied to all low-resolution images to generated the SR-GAN enhanced high-resolution image (SR-HR). These enhanced images (SR-HR) are segmented into macropores, solids, and microporous media, from which a macropore network model (macro-PNM) is extracted. For the sub-resolution microporous media, we employ high-resolution SEM image (0.1 $\mu\text{m}/\text{pixel}$) to characterize micropore structures, which inform the generation of 3D microporous media and extraction of a micropore network model (micro-PNM). The final multi-scale pore network model is created by integrating both macro- and micro-PNMs, effectively capturing the heterogeneity of the rock sample and overcoming the resolution limitations of conventional imaging approaches. The proposed method is validated by comparing the permeability predicted from the multi-scale PNM with experimental measurements, showing excellent agreement and demonstrating its ability to accurately represent complex pore structures.

The novelty of this study lies in its synergistic combination of super-resolution reconstruction, MPS-based 3D pore structure modeling, and multi-scale PNM generation. By bridging the gap between large-FOV imaging and fine-scale structural characterization, this approach provides a powerful framework for digital rock physics, significantly improving reservoir characterization and fluid flow predictions in

heterogeneous formations. The integrated workflow not only addresses current imaging limitations but also opens new avenues for understanding multi-scale transport phenomena in porous media.

Keywords: multi-scale pore network modeling, Super-Resolution, SR-GAN, semi-stochastic pore volume reconstruction

Introduction

Pore-scale investigations of fluid flow in porous media play a pivotal role in advancing energy and environmental sciences, with applications spanning CO₂ geological sequestration, groundwater remediation, hydrocarbon recovery, underground energy storage, and geothermal system assessments (Silin and Patzek, 2006; Dong and Blunt, 2009; Fatt, 1956; Bryntesson, 2002; Al-Kharusi and Blunt, 2007; Al-Raoush et al., 2003; Gao et al., 2012; Jivkov and Xiong, 2014). These studies rely on two main stages: (1) pore structure imaging – advanced techniques such as micro-CT scanning and digital rock physics enable high-resolution characterization of pore networks (Molins, 2015; Molins et al., 2012; Frank et al., 2018; Siena et al., 2015), and (2) fluid flow simulation – numerical methods, including lattice Boltzmann (LB) and pore-network modeling, simulate multiphase flow behavior within these structures (Raeini et al., 2012; Blunt et al., 2002; Joekar-Niasar and Hassanizadeh, 2012; Reeves and Celia, 1996). However, each stage presents significant challenges that limit the result accuracy and applicability.

Micro X-ray computed tomography (micro-CT) and scanning electron microscopy (SEM) are two primary techniques for visualizing the internal pore structures of rock samples (Dunsmuir et al., 1991; Bakke and Øren, 1997; Wildenschild et al., 2002; Arns et al., 2007; Blunt et al., 2013; Wang and Sun, 2022; Wang et al., 2021; Wang, 2023; Wang and Sun, 2021). Micro-CT enables non-destructive 3D imaging of pore networks, offering insights into pore connectivity and spatial distribution. However, its effectiveness is limited by the inherent trade-off between resolution and field of view — higher resolutions require smaller samples, potentially compromising representativeness in heterogeneous rocks (Bakke and Øren, 1997; Bryant and Blunt, 1992; Sahimi and Tahmasebi, 2021).

In contrast, SEM provides high-resolution 2D images, revealing fine-scale pore geometries and surface features (Wang et al., 2018a; Tong et al., 2017; Yang et al., 2010; Park et al., 2003; Chang et al., 2004). Yet, its inability to capture 3D connectivity restricts its utility for analyzing fluid transport pathways. Recent advances in imaging devices and 3D reconstruction techniques aim to bridge this gap, though challenges persist in fully characterization of multi-scale pore systems (Ahuja et al., 2022; Wang and Rahman, 2023; Zannota et al., 2023; Wang et al., 2018b; Bai and Berezovsky, 2020; Gerke et al., 2015; Tao and Muller, 2016; Liu et al., 2023).

This methodology requires a collection of rock sample images at multiple resolutions (minimum two images per resolution level), obtainable through various imaging techniques such as optical microscopy, micro-CT, and SEM. The workflow begins with image registration algorithms aligning images of different resolutions to create matched pairs. These registered image pairs serve as training data for a Convolutional Neural Network (CNN) model, providing crucial information about resolution-dependent image degradation patterns. Early studies by Wang et al. (2018b) applied CNN-based Super Resolution to digital rocks to boost the resolution of the rock images. Their work utilized co-registered micro-CT and SEM images to train the CNN, which then reconstructed high-resolution micro-CT volumes through voxel-by-voxel processing. Subsequent methodological improvements introduced patch-based processing to accelerate reconstruction, replacing the computationally intensive voxel-level approach. Recent advancements have further optimized the process through implementation of pre-trained CNN architectures, which significantly reduce training time while maintaining reconstruction accuracy. (Umehara et al., 2018; Park et al., 2018).

However, sometimes paired images are not always available. Generative adversarial networks (GAN) have been applied in Super Resolution and do not require alignment of training images (Ledig et al., 2017).

Super-Resolution Generative Adversarial Networks (SR-GANs) have emerged as a powerful deep learning approach for enhancing the spatial resolution of low-quality images while preserving critical structural details. Introduced by [Ledig et al. \(2017\)](#) in their seminal work "Photo-Realistic Single Image Super-Resolution Using a Generative Adversarial Network," SR-GANs leverage adversarial training to generate high-resolution (HR) images from low-resolution (LR) inputs, outperforming traditional interpolation and CNN-based methods. By integrating perceptual loss functions and adversarial training, SR-GANs not only enhance pixel-level accuracy but also recover high-frequency geological features that are often lost in conventional upscaling methods.

After successfully visualizing rock pore structures, fluid flow simulation within these complex porous media remains computationally intensive. To overcome this challenge, pore network modeling has emerged as a highly efficient approach. By abstracting the intricate pore geometries and topological connections into simplified networks of pores (nodes) and throats (connections), these models dramatically reduce computational requirements while preserving the essential flow characteristics of the porous medium. This simplification enables practical multiphase flow simulation processes that would otherwise be prohibitively expensive to model using direct numerical approaches on the high-resolution pore structure images. The pore network approach thus provides an optimal balance between computational efficiency and physical accuracy for studying fluid transportation in porous rocks. ([Raeini et al., 2012](#); [Blunt et al., 2002](#); [Joekar-Niasar and Hassanizadeh, 2012](#); [Yuan et al., 2016](#); [Wang and Rahman, 2023](#)).

The field currently employs two primary pore network extraction methods: the medial axis approach ([Jiang et al., 2007](#); [Ma et al., 2014](#)) and the maximum ball method ([Silin and Patzek, 2006](#)). Both methods suffer from inherent limitations in precisely defining pore-throat geometries, leading to potential measurement subjectivity. The medial axis method creates a simplified pore structure skeleton by tracing central pathways, but its performance degrades significantly when encountering surface irregularities ([Dong and Blunt, 2009](#); [Silin et al., 2003](#)). The maximum ball algorithm, while effective for basic pore space representation through iterative sphere placement, struggles to accurately capture topological complexity in intricate pore networks ([Dong and Blunt, 2009](#)).

In this study we used the medial axis-maximal ball method ([Yi et al., 2017](#)) to generate pore network models that integrate the strengths of both the medial axis ([Jiang et al., 2007](#); [Ma et al., 2014](#)) and the maximum ball method ([Silin and Patzek, 2006](#)). By generating maximal balls along the medial axis and implementing refined pore/throat definitions, this technique overcomes previous ambiguities in pore space characterization.

There are two key enhancements to conventional macro- and micro-PNM generation ([Jiang et al., 2013](#)). First, we apply Super Resolution techniques to improve micro-CT image quality, enabling upscaling of PNMs while improving the spatial accuracy of both macropore and micropore distributions. Second, we generate micro-PNM based on SEM imaging data rather than micro-CT, achieving superior resolution for micropore structures while maintaining computational efficiency. This dual approach significantly advances the precision and scalability of multi-scale pore network modeling.

Materials and Standard Properties

This study utilized an open-source Indiana limestone sample from the Digital Porous Media repository <https://digitalporousmedia.org/published-datasets/>. The 6 mm diameter Indiana limestone core plug was scanned at a high resolution of 2.68 $\mu\text{m}/\text{voxel}$, clearly resolving three phases: (1) macropores, (2) solids, and (3) microporous media. However, the micro-CT resolution proved insufficient for characterizing the inner structure of microporous media. To overcome this limitation, high-resolution SEM imaging (0.1 $\times$ 0.1 $\mu\text{m}/\text{pixel}$) was employed to capture the inner structure of microporous media, enabling comprehensive analysis of the dual-porosity system. The grayscale intensity variations in the micro-CT images effectively distinguished the different pore structures within the Indiana limestone sample.

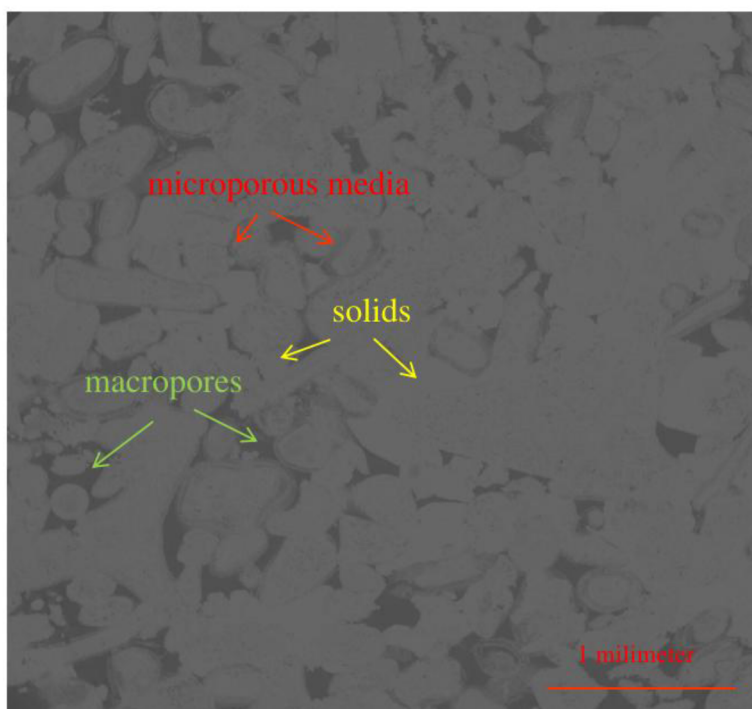


Figure 1—A representative micro-CT image used in this study.

Methodology

This study presents a novel multi-scale pore network modeling approach that integrates Super Resolution and multi-scale pore network modeling. Because there is no paired different resolution images, in this study, we use down sampled images (LR) as training image, and the original high resolution images (HR) as evaluation data. The workflow begins with high-resolution ($2.68 \mu\text{m}/\text{voxel}$) micro-CT imaging (HR) of a 6 mm core plug, which is down sampled to create low-resolution ($10.72 \mu\text{m}/\text{voxel}$) training images (SR) for a Super-Resolution Generative Adversarial Network (SR-GAN). The trained SR-GAN model then applied to all low-resolution images to generate the SR-GAN enhanced high-resolution image (SR-HR). These enhanced images (SR-HR) are segmented into macropores, solids, and microporous media, from which a macropore network model (macro-PNM) is extracted. For the sub-resolution microporous media, we employ high-resolution SEM image ($0.1 \mu\text{m}/\text{pixel}$) to characterize micropore structures, which inform the generation of 3D microporous media and extraction of a micropore network model (micro-PNM). The final multi-scale pore network model is created by integrating both macro- and micro-PNMs, effectively capturing the heterogeneity of the rock sample and overcoming the resolution limitations of conventional imaging approaches.

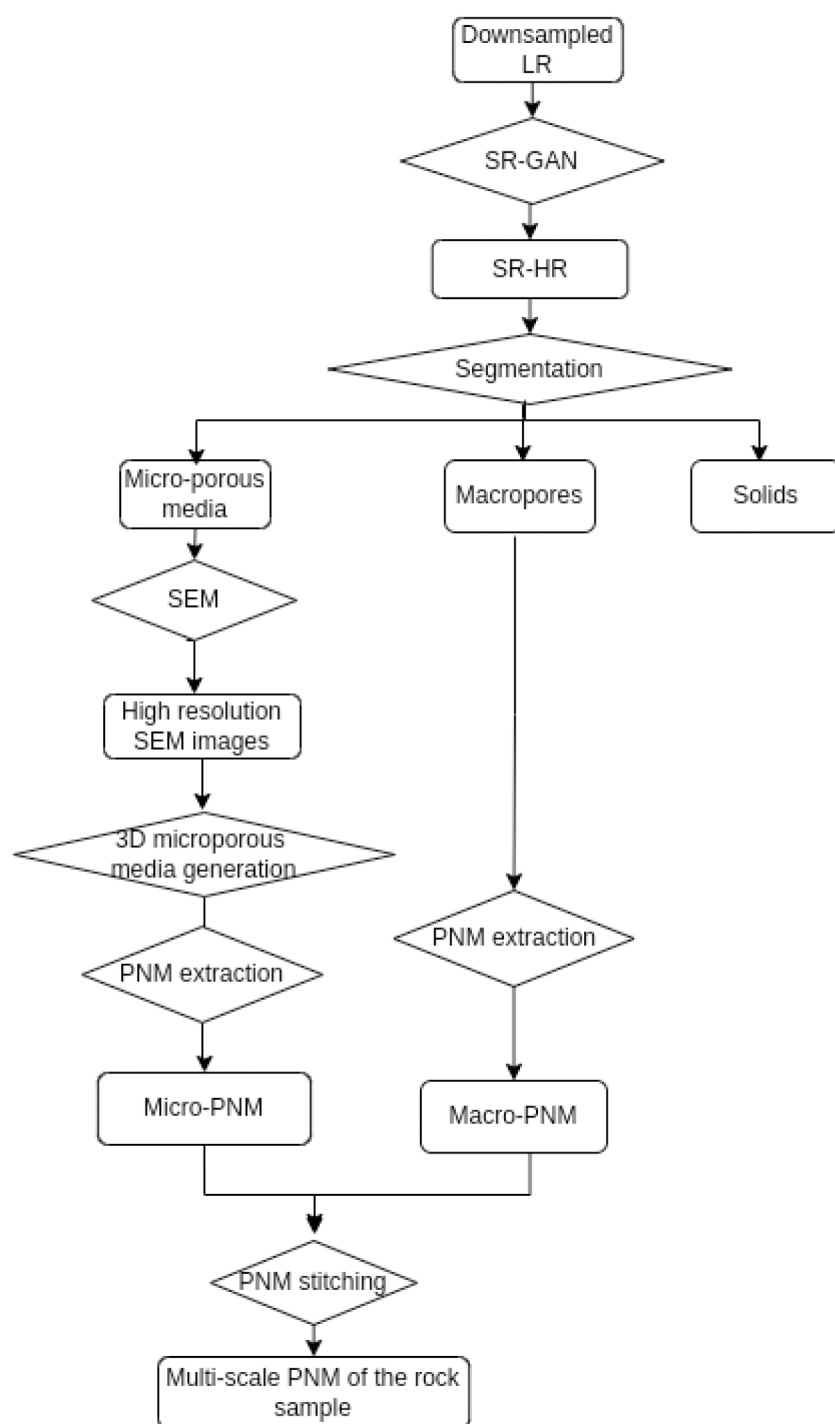


Figure 2—Workflow of the multi-scale pore network modeling.

Procedure of SR-GAN

To maintain computational efficiency while capturing essential geological features, we adopted an image patching strategy with 256×256 -pixel image patches extracted using a sliding window approach (40-pixel step size in x/y directions). This configuration ensured comprehensive coverage of porous media textures while optimizing memory usage and minimizing edge artifacts. SR-GAN framework enhanced micro-CT image resolution through an adversarial training process between generator and discriminator networks. The training protocol utilized down sampled low-resolution images (LR) as inputs while maintaining original HR images as ground truth targets for quality assessment. The generator network learned to produce high-

resolution outputs (SR-HR) by optimizing a composite loss function incorporating: (1) pixel-wise mean squared error loss for structural fidelity, (2) perceptual loss based on VGG network features for texture preservation, and (3) adversarial loss for photorealistic quality. The repeated training process helped the generator successfully rebuild small pore details and natural textures from low-quality images. The resulting high-quality images were both accurate and looked as good as original high-resolution images. Fig. 3 shows the procedure of SR-GAN in this study

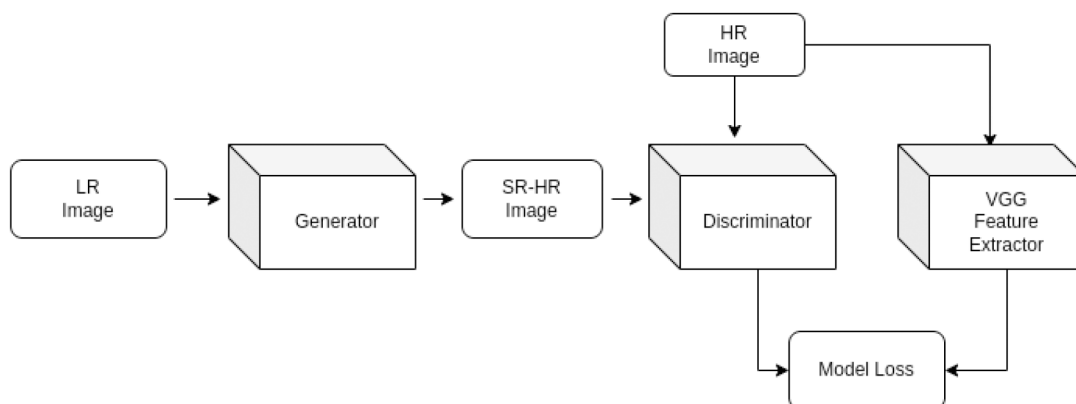


Figure 3—Working procedure of SR-GAN

Pore network model generation

This study developed a three-stage workflow to construct a multi-scale pore network model (PNM) capturing the 3D pore structure heterogeneity of the rock sample. The methodology involved generating separate PNMs for macropores and microporous media before integrating them into a unified model (Fig. 4).

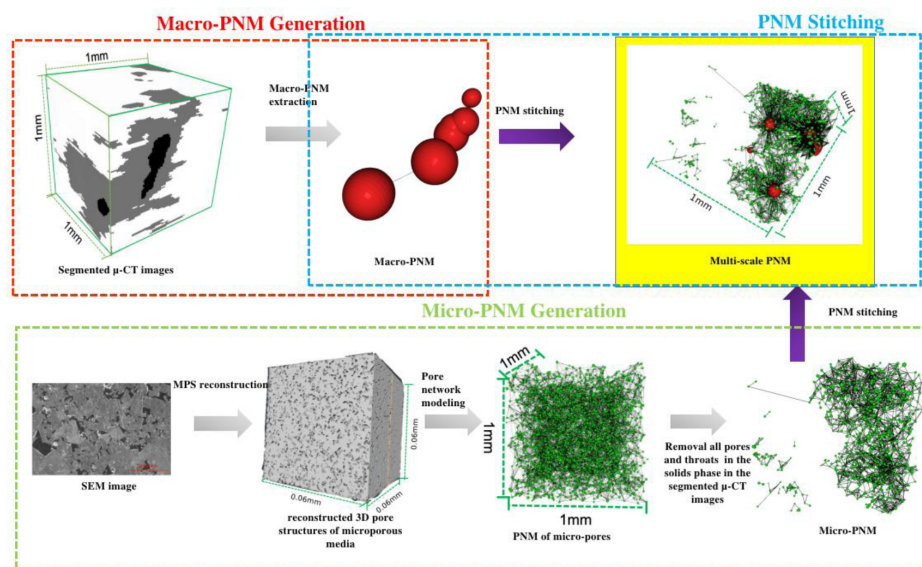


Figure 4—Illustration of the multi-scale pore network modeling process. The process starts with the generation of both macro- and micro-PNMs and finishes with the creation of a multi-scale model (yellow frame).

First, the macropore network model (macro-PNM) was directly extracted from the SR-GAN-generated micro-CT images (SR-HR) using the medial axis-maximal ball algorithm (Yi et al., 2017). Second, a micropore network model (micro-PNM) is generated through three key steps: (1) high-resolution SEM imaging ($0.1 \times 0.1 \mu\text{m}/\text{pixel}$) of microporous media, (2) 3D pore structure reconstruction using multiple-point statistics applied to SEM data, and (3) network extraction via the medial axis-maximal ball

method. This innovative combination of imaging and modeling techniques overcame micro-CT resolution limitations while maintaining computational efficiency. Finally, the macro-PNM and micro-PNM were computationally integrated to form a multi-scale PNM, effectively representing the complex pore system spanning multiple length scales within the rock sample. This integrated approach provided a comprehensive 3D characterization of the dual-porosity system while balancing resolution requirements with computational practicality.

Macro-PNM generation. The macro-PNM was directly extracted from the segmented SR-GAN enhanced micro-CT images using the medial axis-maximal ball method (Yi et al., 2017). The medial axis-maximal ball method results in the generation of maximum balls along the medial axis, followed by the classification of the balls into either pore balls or throat balls depending on the size and location. Pore balls are defined as either: (1) end-point balls, (2) conjunction balls (a conjunction ball here means a maximum ball has more than two neighbor balls), or (3) local maximum balls. A local maximum ball is a ball larger in size than its connected balls. The other balls on the medial axis are classified as throat balls.

Micro-PNM generation. High-resolution SEM imaging ($0.1\ \mu\text{m}/\text{pixel}$) of the microporous media (Fig. 5a) revealed intricate microporous networks - features entirely undetectable at micro-CT resolutions. However, the inherent 2D nature of SEM imaging precludes direct assessment of pore connectivity and fluid flow properties which are essential for accurate pore network modeling. To address this, we implemented a Multiple-Point Statistics (MPS) method (Strebelle, 2002) to reconstruct representative 3D pore structures from segmented SEM images, generating a $1000 \times 1000 \times 1000$ voxel volume that effectively captures the microporous media's characteristic pore structure while remaining computational efficiency. From this 3D reconstruction, we extracted a micro-PNM using the medial axis-maximal ball method (Fig. 5), successfully transforming 2D SEM image into a 3D pore connectivity framework.

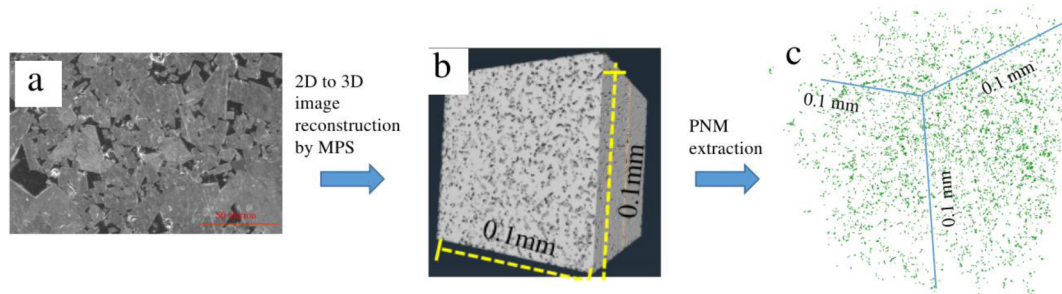


Figure 5—The workflow of micro-PNM generation. (a) SEM image of microporous media at a resolution of $0.1 \times 0.1\ \mu\text{m}^2/\text{pixel}$. (b) Reconstructed porous medium based on the SEM image in (a) using MPS at $1000 \times 1000 \times 1000$ voxels. (c) PNM extracted from the porous medium.

PNM stitching. The multi-scale pore network model was constructed by integrating macro-PNM and micro-PNM into one integrated pore system. The stitching procedure involves three critical steps: First, the macro- and micro-PNMs are spatially aligned, requiring resolution matching through image interpolation when necessary to ensure dimensional consistency. Second, topological conflicts are resolved by removing micropores that geometrically overlap with macropores. Third, new connections were made between the remaining micro pores and nearby macro pores while keeping their original sizes. This stitching methodology (shown in Fig. 6) creates a seamless multi-scale pore network that retains the characteristics of both macropores and micropores while accurately representing their spatial interconnections. The resulting multi-scale model overcomes the limitations of single-scale approaches by simultaneously capturing: (1) the interconnected macropore network governing bulk fluid flow, and (2) the microporous domains that contribute to storage capacity and localized transport phenomena.

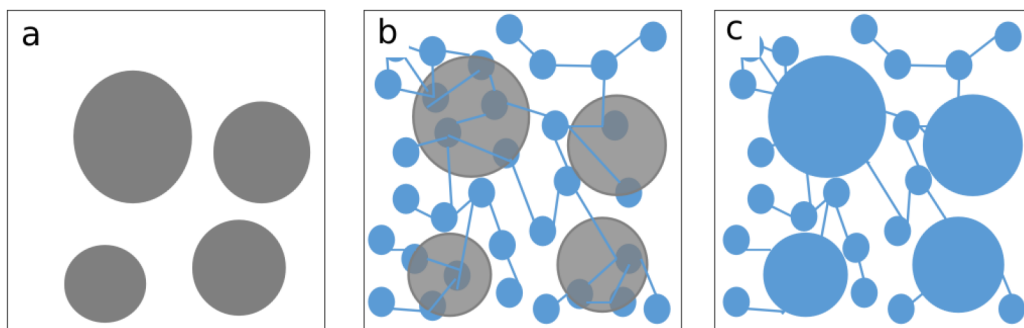


Figure 6—Macro- and micro-PNMs stitching process. (a) Schematic of a macro-PNM that consists of four macropores. (b) Schematic of overlapping macro- and micro-PNMs. (c) Schematic of multi-scale PNM after removal of the micropores covered by the macropores. This also shows the connections between the remaining micropore neighbors and the macropores.

Results and discussion

Super Resolution results and discussion

Fig. 7 presents the Super Resolution (SR) results obtained using the SR-GAN, along with a comparative analysis of segmented images derived from the target low-resolution image (LR), the SR-GAN-enhanced image (SR-HR), and the reference high-resolution image (HR). The results demonstrate that our SR-GAN model effectively reconstructs high-fidelity micro-CT images, recovering intricate structural details that are otherwise indistinguishable in the low-resolution input.

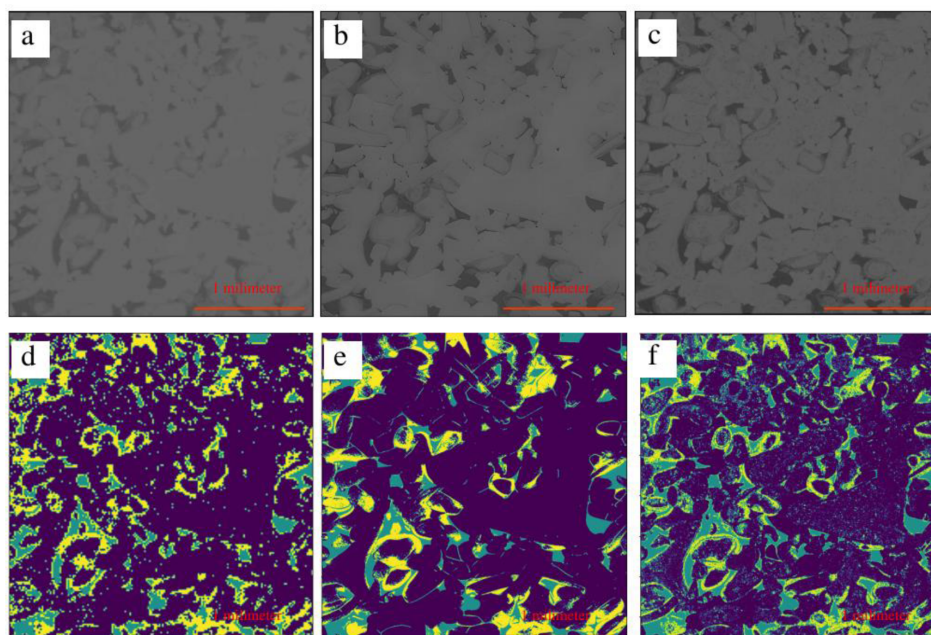


Figure 7—The results of SR-GAN and image segmentation. (a) is the down sampled micro-CT image (LR), the target image. (b) is the SR-GAN improved image (SR-HR). (c) is the reference image (HR). (d) is the segmented image of (a), dark blue represents solid, green represents macropores and yellow represents microporous media. (e) is the segmented image of (b). (f) is the segmented image of (c).

A key observation from the segmented image comparison is the superior delineation of phase boundaries in the SR-GAN-reconstructed image, closely matching the reference image. In contrast, the low-resolution target image fails to preserve fine edges and microstructural features, leading to blurred and inaccurate segmentation. This improvement highlights the critical role of our SR-GAN in enhancing image quality for pore-scale characterization.

The proportion of each phase was calculated with macropore pixels constitute 8.1% of the entire image pixels and microporous medium pixels account for 17.6% of the entire image pixels. These values align closely with the corresponding reference images of HR, which calculated macropore pixels as accounting for 8.2% and microporous medium pixels accounting for 18.8% of the entire image.

The evaluation of the pixel-wise accuracy of the segmentation results used the intersection over union (IoU) metric. IoU value of 0.95 is achieved for the macropores and 0.90 for the micro-porous media. The high level of accuracy highlights the capability of the SR-GAN model to accurately capture and delineate phase features within the micro-CT images and indicates the model's capacity to recover microporous media which were not clearly discernible in the original inch-plug micro-CT images.

The accurate segmentation of micro-CT images provides the distribution of each phase within the rock sample which facilitates further detailed analysis of pore structures and fluid flow behavior within the sample. However, the details of porous structures cannot be observed using micro-CT because of the inherent resolution limitation. Thus, we discuss how to overcome this issue by integrating SEM and micro-CT images and use SEM to visualize the micropores with nanometer-scale resolution.

Multi-scale pore network modeling results and discussion

Fig. 8 presents the multi-scale PNM. The red balls represent macropores which were directly extracted from the SR-GAN improved micro-CT images. The green balls represent micropores which were generated using an SEM training image. The domain of the micropores depends on the microporous media phase in the segmented improved micro-CT images. These results show both the visible macropores and finer details of the micropores and provide a comprehensive representation of the complex dual porosity system within the rock sample.

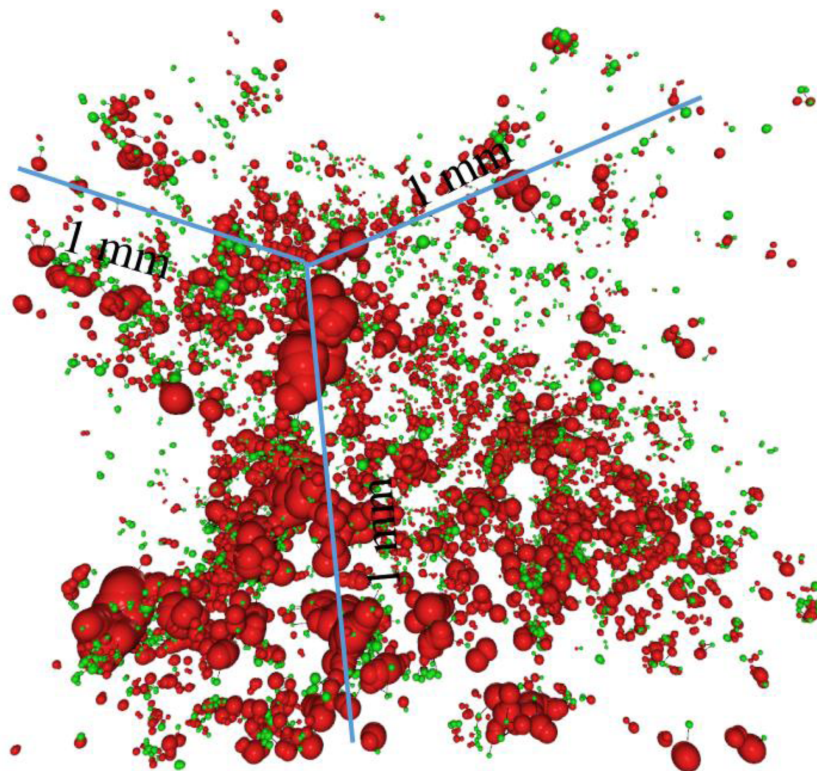


Figure 8—Multi-scale PNM at $1 \times 1 \times 1 \text{ mm}^3$. Red balls represent macropores and green balls represent micropores.

The multi-scale PNM extracted 5646 macropores with an average coordination number of 6.8. The average coordination number of the micropores is 5.5. Both coordination number indicate a high number of

interconnections between micropores and macropores. The permeability of the generated multi-scale PNM was estimated using a single-phase flow simulation. The results show that the permeability of the PNM is 14.7 mD, which is very similar to the value of 16-20 mD measured in the laboratory on the Indiana limestone. This close alignment highlights the potential of this method.

Conclusions

A multi-scale pore network modeling method has been developed by integrating a Super Resolution technique to accurately visualize solids, macropores and volumes containing micro-porosity and a Multiple Point Statistics to generate 3D porous media of the microporosity from 2D segmented SEM images. This method was developed and successfully applied in an open-source Indiana limestone. Key conclusions are:

1. The SR-GAN architecture adequately improved the low-resolution micro-CT images (LR at a resolution of $10.72 \mu\text{m}^3/\text{voxel}$). The reconstructed images reproduced the phase distributions within the core plug sample which could not be clearly delineated in the original images. In addition, the high IoU values of $>90\%$ obtained for macropores and microporous media highlight the capacity of the model to segment the low-resolution micro-CT images.
2. The medial axis-maximal ball method is an enhanced version of both the medial axis and maximal ball methods. The pore network model generated using this method prevents the production of superfluous 'micropores' and ensures all throats are located within the hydraulic restrictions. Consequently, the medial axis-maximal ball method successfully retains both the topological and morphological features of pore structures.
3. The novel code developed for PNM extraction and PNM stitching is computationally efficient and significantly reduces processing time and addresses a commonly found limitation in many commercially available software.
4. The multi-scale PNM generated using a two-step strategy of macro-and micro-PNMs generated separately considers effectively sample size, resolution and computational demand.

Declarations

Conflict of Interest: The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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